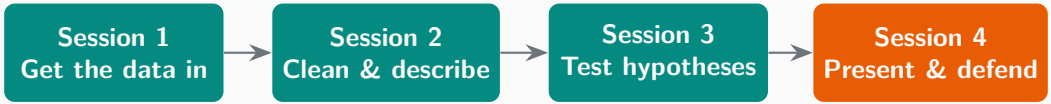
Python for Data Science

Session 4 — Presenting and Defending Your Answer

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Three sessions of tools; today you show what you can do with them



Where we left off: each research question has an answer, a test behind it, and a limit. Education adds almost nothing once repayment is known; limits partly reflect the bank's choice; late months predict default.

Today: every group presents, every student answers questions on the code, and then we ask why groups disagree.

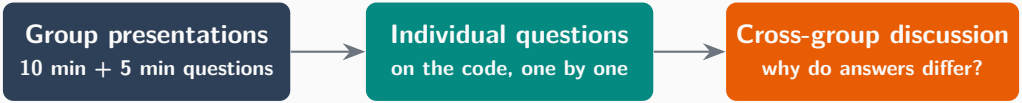
Three cleaning choices $\times$ one definition $\times$ one modelling choice

24

defensible ways to analyse the same 30,000 clients

Do they give the same answer? We will find out after the presentations.

Today runs in three parts



Groups working on the same research question present one after the other, so we can compare their answers directly.

Part 1: How you are assessed

The same two criteria as on day one

Two criteria, announced in Session 1, now made concrete

	Insufficient	Good	Excellent
Command of the code	cannot say what a line does	explains each line	explains, modifies on the spot, and predicts the effect of a change
Critical thinking	reports p-values only	reports size and significance	states what the method can and cannot answer, and why

Both criteria are assessed for the group (presentation) and for each student (questions).

The presentation: answer first, then how you got there

Min.	Content	Slides
1	The question and your answer, in one sentence	1
2	Cleaning decisions that matter for your question	1
3	One plot that shows the pattern	1
3	One test: result, effect size, interpretation	1–2
1	The main limit of your answer	1

Titles state findings (“Defaulters had 48k lower limits”), not topics (“Results”), as on every slide of this course.

Individual questions come in three kinds

Explain

"What does this line do?"

```
(df[cols] >=
1).sum(axis=1)
```

Say what goes in, what comes out, and why you need it.

Justify

"Why did you do it this way?"

Why Welch and not the standard t-test? Why set codes 0, 5, 6 to missing?

Point to your cleaning log.

Predict

"What happens if. . . ?"

You drop `late_months` from the regression. What happens to the limit coefficient?

Reason first, then run it.

When you do not know the answer, reason out loud

Weak answers

"I don't know."

"The AI wrote that part."

"It's significant, so it's important."

Strong answers

"I'm not sure, but `axis=1` should mean across columns, so. . . "

"Let me print the first rows to check."

"It's significant, but the effect is small: $V = 0.06$."

Why it matters: we assess how you think, not whether you memorised the output.

Part 2: Group presentations

10 minutes each, then questions

Running order

Slot	Research question	Group	Time
1	**RQ1**: education and default		
2	**RQ1**: education and default		
3	**RQ2**: credit limits		
4	**RQ2**: credit limits		
5	**RQ3**: repayment history		
6	**RQ3**: repayment history		

While another group presents, note one point where your answer differs from theirs.

Part 3: Same data, different answers?

Where the groups' paths split

Every group took the same four forks, sometimes in different directions

Fork	Options	Session
Undocumented education codes	missing / merge into "others" / drop rows	2
Duplicate rows	keep / drop	2
What counts as a "late" month	1+ months late / 2+ months late	2
Control variables	none / limit, age, sex, education	3

$3 \times 2 \times 2 \times 2 = 24$ paths. We ran all of them with one loop over a cleaning **function**, the tools of Session 1 at full scale.

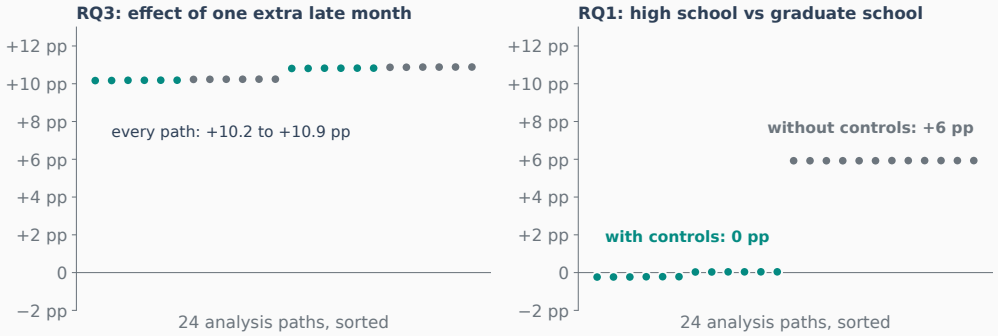
Twenty-four analyses in a dozen lines of code

```
import itertools

results = []
for edu, dup, thr, ctrl in itertools.product(
        ["missing", "others", "drop"], [False, True],
        [1, 2], [False, True]):
    df = clean(raw, edu_unknown=edu, drop_duplicates=dup,
               late_threshold=thr)
    formula = "default ~ late_months + limit10k"
    if ctrl:
        formula += " + age + C(sex) + C(education)"
    res = smf.ols(formula, data=df).fit(cov_type="HC1")
    results.append(res.params["late_months"])
```

Full code in `session4_multiverse.ipynb`. Try it on your own group's choices.

Cleaning choices barely matter; the choice of controls decides RQ1



Each dot is one of the 24 paths. **Teal**: with controls. **Gray**: without. The RQ3 answer survives every path. The RQ1 answer is 6 pp or 0 pp depending on a single choice.

Which fork moved which answer

	RQ1 high school vs graduate	RQ2 limit, per 100k	RQ3 per late month
Unknown education codes	< 0.1 pp	< 0.1 pp	< 0.1 pp
Duplicates	< 0.1 pp	< 0.1 pp	< 0.1 pp
"Late" = 1+ or 2+ months	< 0.3 pp	< 0.1 pp	10.2 vs 10.9 pp
Controls: none or all	**5.9 vs 0.0 pp**	**−4.9 vs −2.1 pp**	10.2 vs 10.2 pp

The cleaning decisions you worried about in Session 2 hardly moved anything. The question "compared with whom?" moved two of the three answers.

Two groups can both be right if they answer different questions

"High-school clients default more"

True: 25% against 19%. The right answer if the bank only knows a client's education, e.g. for a new applicant.

"Education does not matter"

Also true: 0 pp once repayment is known. The right answer for an existing client with six months of history.

Why it matters: a disagreement between groups is often a disagreement about the question. Find it before you argue about the numbers.

Discussion: three questions for the whole class

1. Which group's answer to your research question is the **most convincing**, and why? (Not: which has the smallest p-value.)

2. Which **limit** did every group share, whatever their choices?

3. If the bank asked you for **one sentence** on who is likely to default, what would you tell it, and what would you warn it about?

Where we started, and where we arrived

Could the bank have seen it coming?

Four sessions, one question, answered step by step

	Tools	What we learnt about default
S1	import, inspect	22% defaulted; the data hide undocumented codes
S2	clean, groupby, plots	12% to 70% as late months rise
S3	tests, regression	+10 pp per late month; education and limits matter less than they seem
S4	robustness	RQ3 survives every path; RQ1 depends on the question asked

The answer: the bank could see who was riskier, not who would default

Yes

Six months of repayment history separate a 12% group from a 70% group.

Partly

$R^2 = 0.16$. Most of what decides default (income, job, other debts) is not in the data.

Here

One bank, Taiwan, 2005, in a credit card crisis. Prediction, not cause.

A good analysis is not the one with the smallest p-value.
It is the one you can defend line by line.